

Scale Effects in GHAR and GNNHAR-IV Volatility Forecasting

Volatility Research Pipeline

June 11, 2026

1 Research Question

We test whether graph structure becomes more valuable as the cross-sectional universe expands. For each universe, the target is 30-day realized volatility and the exogenous option-market channel is 30-day mean implied volatility. The key estimand is the out-of-sample loss reduction $1 - L_{\text{enhanced}}/L_{\text{baseline}}$, reported for MSE and QLIKE.

2 Universe-Level Results

Table 1: Best models and baseline losses by universe

Universe	Assets	Dates	HAR QLIKE	GHAR+IV QLIKE	Best QLIKE model
Dow30	30	1234	0.000821965	0.000784934	GHAR+IV
S&P100	99	1073	0.0033707	0.00323373	GHAR+IV
S&P500	449	1223	0.00413344	0.00402077	GHAR+IV

3 Scale Gains

Table 2: Selected QLIKE gains by universe

Universe	Assets	GHAR over HAR	GHAR+IV over GHAR
Dow30	30	1.32%	3.22%
S&P100	99	0.23%	3.84%
S&P500	449	0.07%	2.66%

4 Interpretation

A positive GHAR-over-HAR gain means the estimated asset graph adds forecasting information beyond own-asset HAR lags. A positive GHAR+IV-over-GHAR gain means the implied-volatility channel contributes information beyond the graph. The GNN rows in the CSV output evaluate whether nonlinear message passing improves on the linear graph benchmark.

5 Artifacts

The machine-readable tables are `scale_summary.csv` and `scale_gains.csv`. The main figures are saved under `figures/`.